Course in Microsystems
University of Minho, Braga, Portugal

Microfluidics

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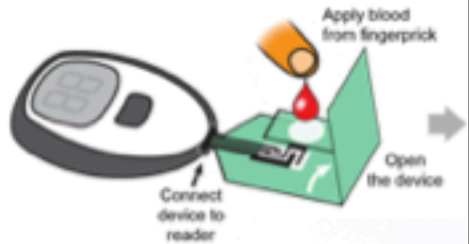
Microfluidics & me

- I started to work on microfluidics 2007 (in my MSc thesis) in Sweden (Chalmers University of Technology)
- … continued with it in my PhD in Sweden (Prof. Owe Orwar)
- … and post doc in US (Prof. George Whitesides at Harvard)
- Over 20 papers and number of patents in microfluidics
- Founded two companies developing microfluidic products.
- … and working also on other aspects of it (e.g. education, standardization)

FLUIKA

fluicell
lab-on-a-tip



Outline for next three classes

- Foundations of Microfluidics
- Making Microfluidics
- Using Microfluidics (Applications)

- At the end of first aclass you will have a little homework (due to next class)

Foundations of Microfluidics

Outline for today

- What is Microfluidics?
- Brief history of Microfluidics?
- Why Microfluidics? Why should we care about it?
- Scaling laws - why “micro” is different?
- Different transport phenomena
  - Flow (Flows on microscale & some practical calculations)
  - Diffusion
  - Flows & transport driven by electricity
    - Electroosmosis
    - Electrophoresis
    - Dielectrophoresis
  - Thermally driven flow (thermophoresis)
- Surfaces
  - Capillary effect
  - Electrowetting
  - Liquid-liquid interfaces and droplets
- Going even smaller - nanofluidics

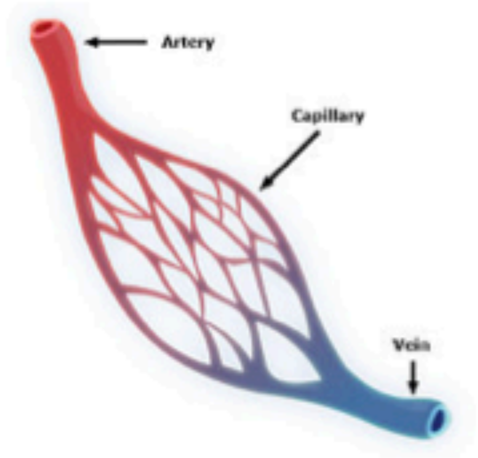
What is Microfluidics?

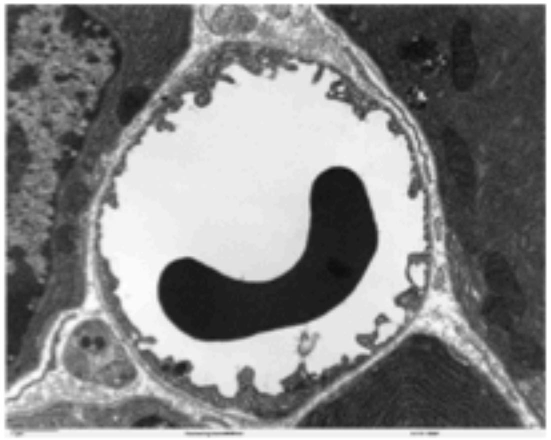
It is the science and technology of systems that process or manipulate small (10^{-9} to 10^{-18} litres) amounts of fluids, using channels with dimensions of tens to hundreds of micrometres

Prof. George M. Whitesides
"The origins and the future of microfluidics"
Nature, 2006

Background & History

First practitioner of Microfluidics - “Mother Nature”





An example: Capillary blood vessel

5-10μm in diameter

Total length in human body:
100’000km (2.5x around the world)

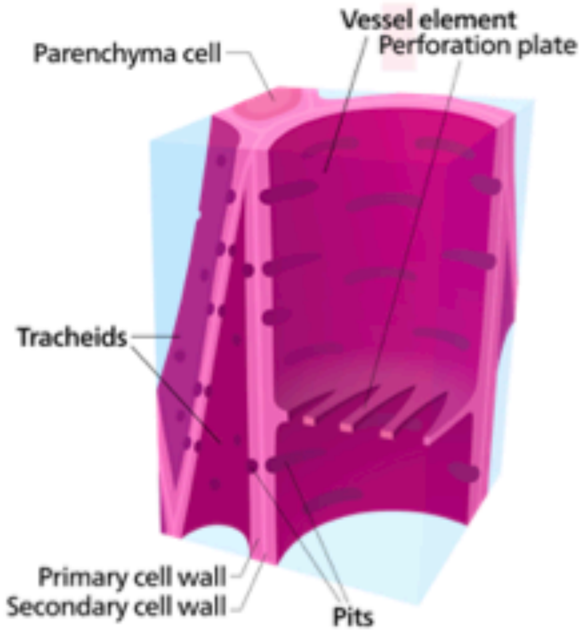
Only: 5.6L of blood
Blood circulation: 3 times every minute

10-100μm from cells

https://en.wikipedia.org/wiki/Capillary

Background & History

Similarly plants have xylem & phloem



Images: Wikipedia

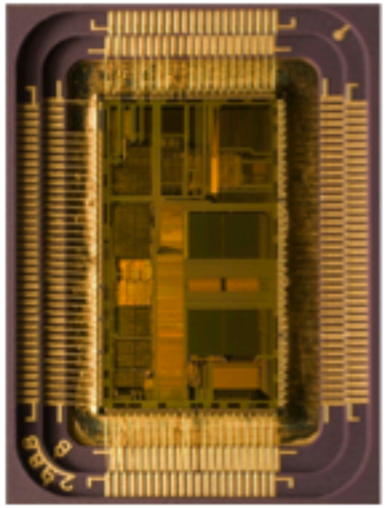
Background & History

Origins of human engineered Microfluidics

Molecular analysis

Biodefence

Molecular- and cell biology

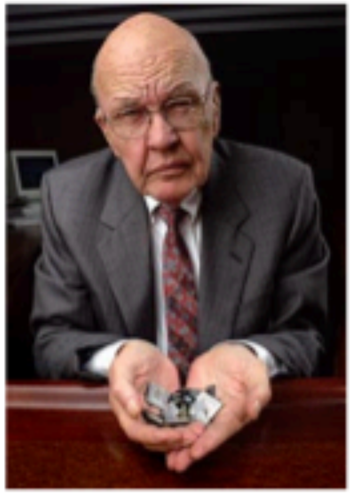


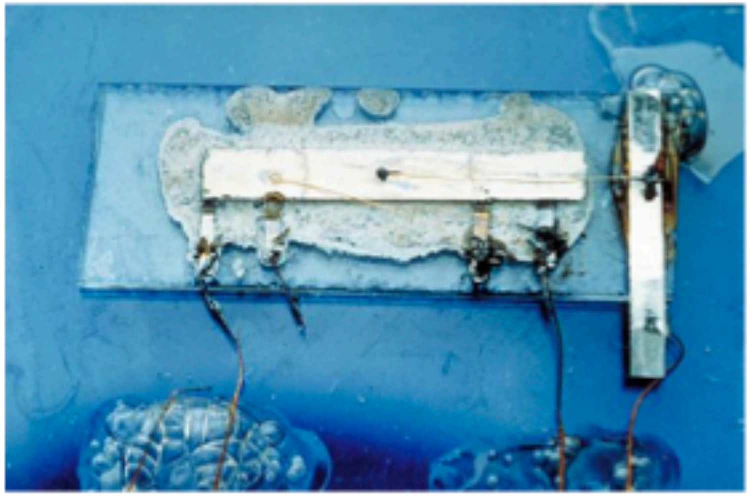
Microelectronics

Images: Wikipedia, http://bkubi.weebly.com/,https://www.hdfgroup.org,

Background & History

Microelectronics - the enabler





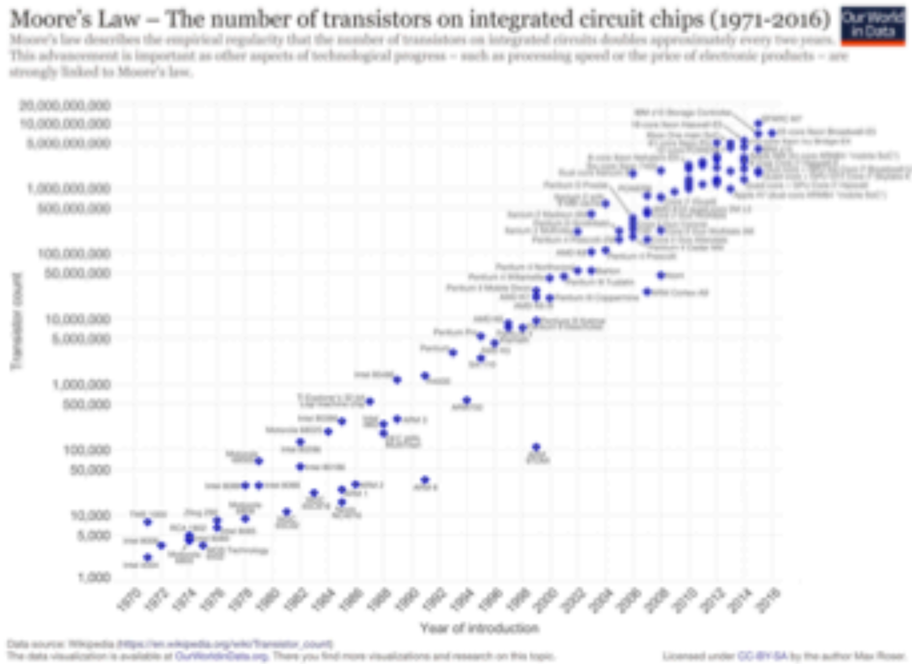
Jack Kilby's & his IC
Solution to "Tyranny of numbers"
Invented in TI in 1958
US Patent 3,138,743
Nobel Price in Physics 2000

Images: Wikipedia

Background & History

Microelectronics - the enabler

Gordon Moore, chemist, businessman and founder of Intel Corporation



Moore's law (1965 revised 1975) doubling components in every 2 years.

Images: Wikipedia

Background & History

Microelectronics - the enabler

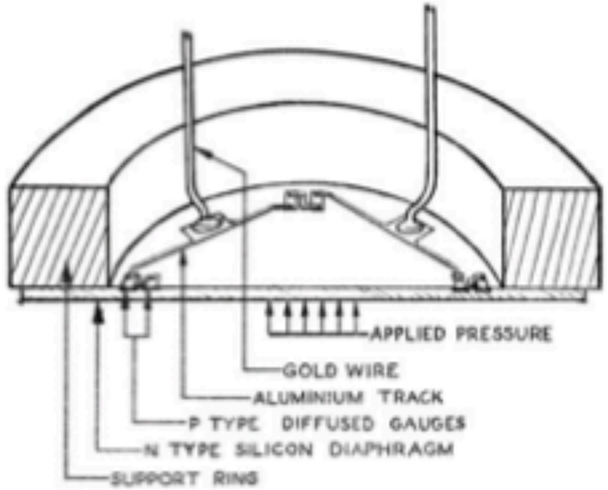
Electronics industry & huge economy behind it have driven miniaturization
- **advanced top-down micro- and nanofabrication**

But it can be also used for other purposes!

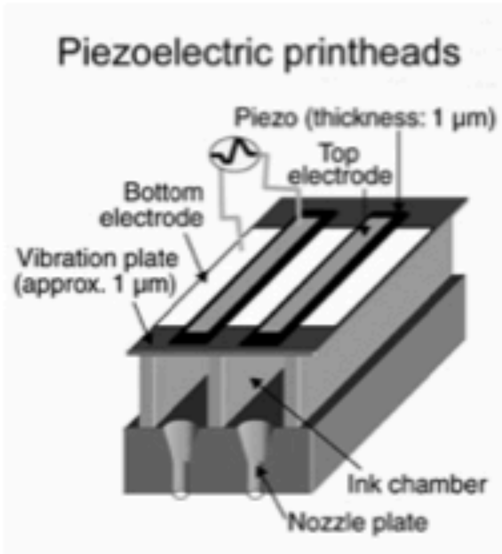
Images: INL

Background & History

MEMS and first fluidic devices



Pressure sensors from 1960s & 70s



Different inkjet technologies

Some ideas originated already from 19th century (Lord Kelvin's patent 1867), 1951 commercial use by Siemens

Images: Bogue, Sensor Review, 2007; https://graphicartsmag.com/articles/2010/02/the-magic-behind-inkjet-printing/

Background & History

Biodefence - worry from the Cold War era & bioterrorism

Science fiction
Star Trek "Tricorder" 1960s

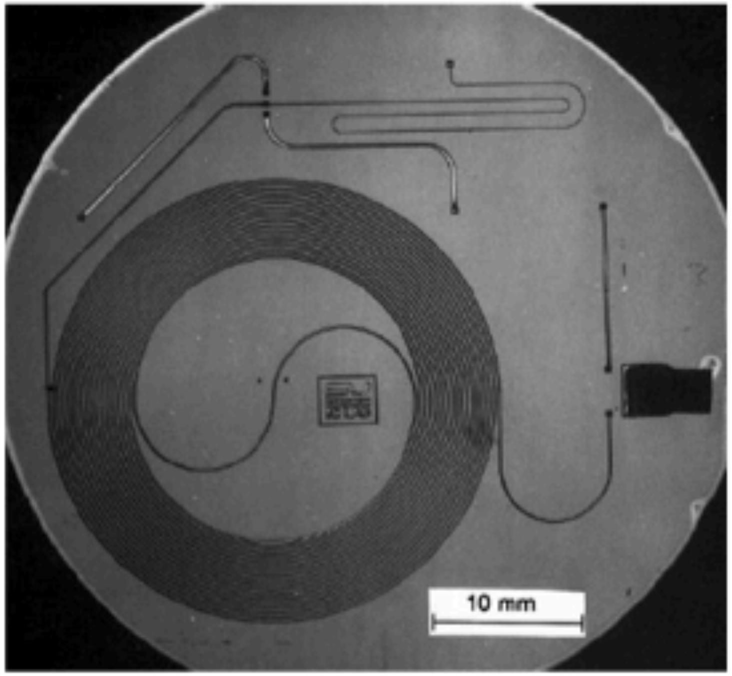
DARPA

Defense Advanced Research Project Agency funded a lot of early projects for technologies enabling portable detection of chemical and biological warfare

Images: thefederationtimes.com, DARPA

Background & History

Molecular analysis



Using MEMS technology for chemistry

Terry et al “A gas chromatography air analyzer fabricated on a silicon wafer” IEEE Transactions on Electron Devices, 1979

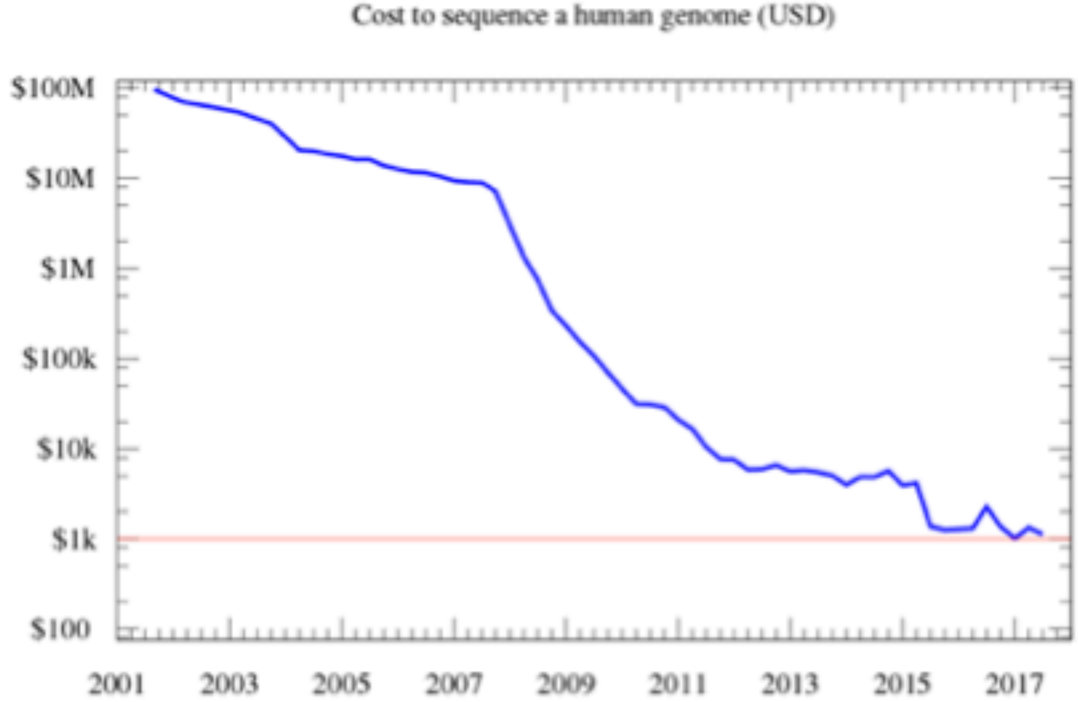
Images: Azzouz et al, Anal Bioanal Chem, 2014, 406:981-994

Background & History

Genomics, sequencing and molecular biology

Frederick Sanger

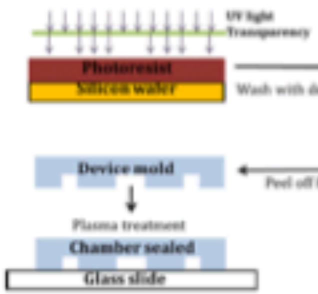
Started: 1970s

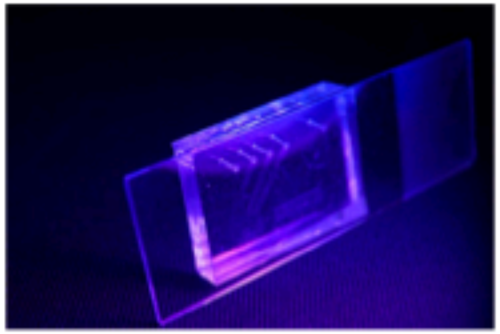


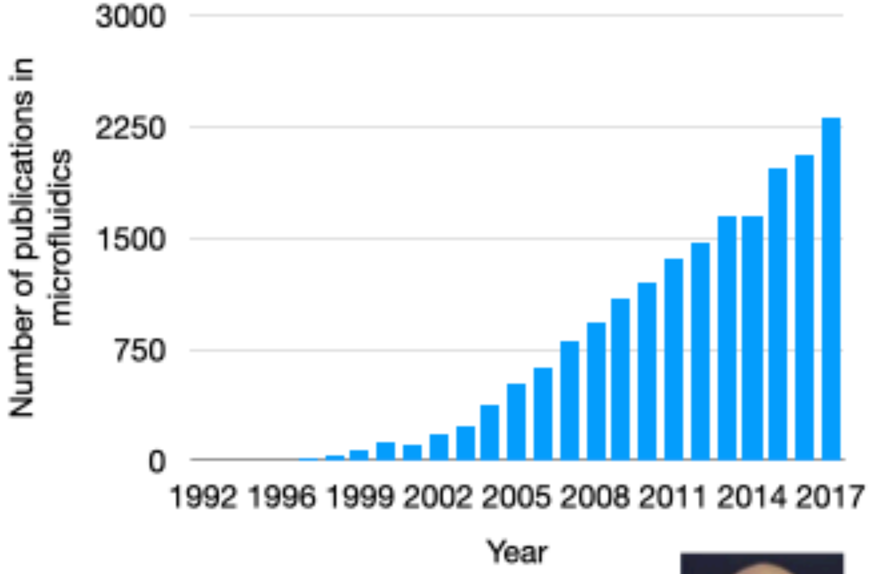
Images: Wikipedia

Background & History

Simplicity & explosion of microfluidic research







Started from Whitesides group
Duffy et al "rapid prototyping of microfluidic systems in poly(dimethylsiloxane)", Analytical Chemistry, 1998 (cited over 5000 times)

Images: Feng et al Chem Comm, 2015; Elveflow

Why Microfluidics? Why to bother?

Reduction …

Amounts of reagents

Amounts of sample

Size and cost of the instrumentation

Time of analysis

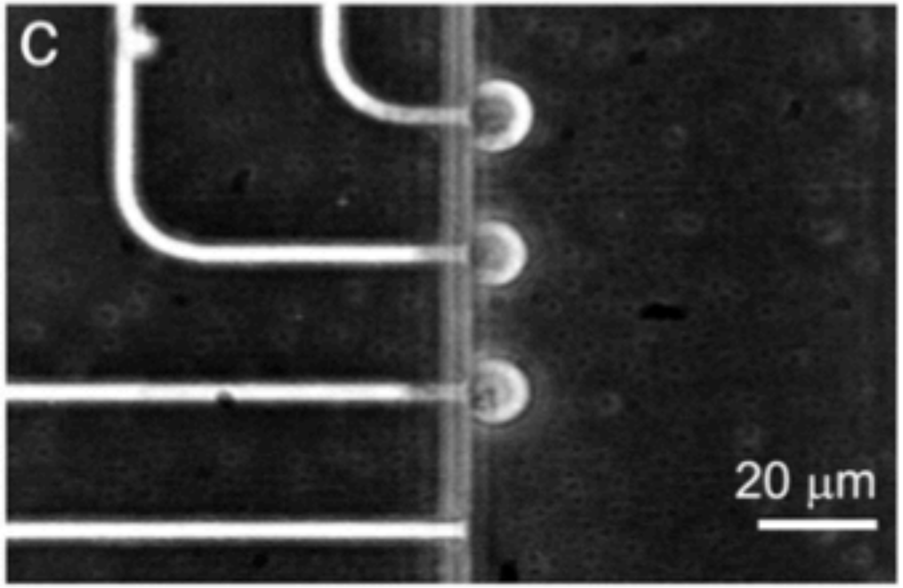
Cost of analysis

Images: Wikipedia

Why Microfluidics? Why to bother?

Fundamentally new possibilities

Different physics & behaviors



Size of cells

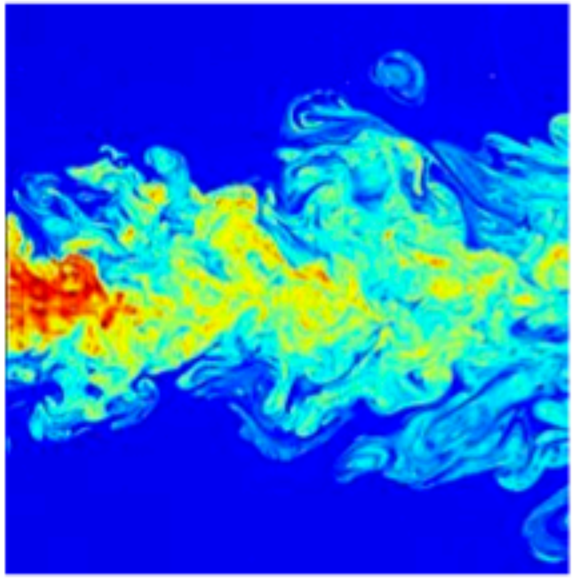
Images: Albert Folch's Lab; Ionescu-Zanetti, PNAS, 2005

Scaling laws - why “micro” is different?

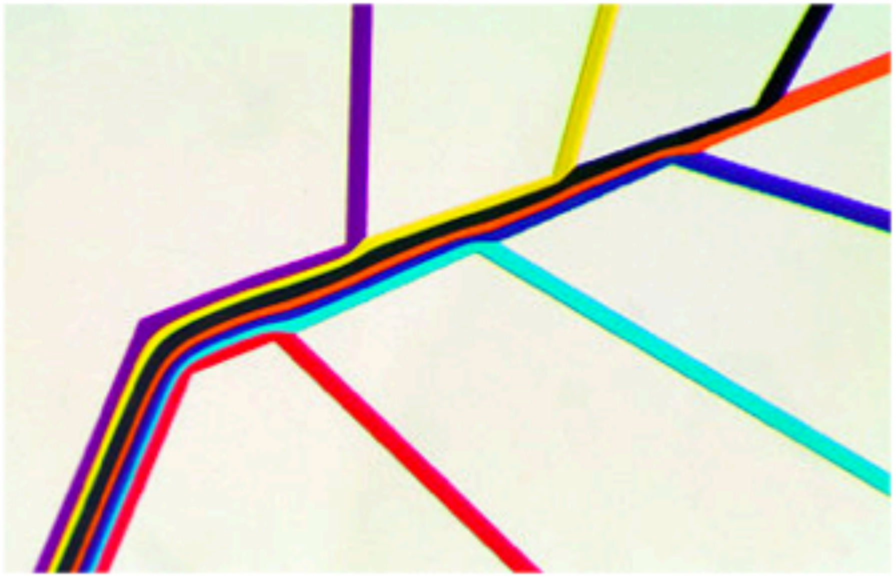
Area	ℓ^2	Eq. (1.1)	Time	ℓ^0	Section 1.2
Volume	ℓ^3	Eq. (1.1)	Velocity	ℓ^1	Section 1.2
Rel. fluctuations	$\ell^{-\frac{1}{2}}$	Exercise 1.3	Hydrostatic pressure	ℓ^1	Eq. (3.3)
Reynolds number *Re*	ℓ^2	Eq. (2.39)	Hydraulic resistance	ℓ^{-4}	Table 4.1
Péclet number	ℓ^2	Eq. (5.53)	Stokes drag	ℓ^1	Eq. (3.128)
Diffusion time	ℓ^2	Eq. (5.10)	Particle diffusion const.	ℓ^{-1}	Eq. (6.49)
Fluid acceleration time	ℓ^2	Eq. (6.25)	Taylor dispersion time	ℓ^{-2}	Eq. (5.71)
Young–Laplace pressure	ℓ^{-1}	Eq. (7.8)	Contact angle	ℓ^0	Eq. (7.14)
Bond number *Bo*	ℓ^2	Eq. (7.30)	Capillary rise height	ℓ^{-1}	Eq. (7.21)
Marangoni force	ℓ^{-1}	Eq. (7.39)	Capillary speed	ℓ^1	Eq. (7.36)
Electric field	ℓ^{-1}	Eq. (8.3a)	EO velocity	ℓ^{-1}	Eq. (9.11)
Ionic mobility	ℓ^{-1}	Eq. (8.15)	EO mobility	ℓ^0	Eq. (9.12)
Debye length	ℓ^0	Eq. (8.26)	EO flow rate	ℓ^1	Eq. (9.35a)
Debye frequency	ℓ^{-1}	Eq. (8.45)	EO pressure	ℓ^{-2}	Eq. (9.35b)
DEP force, particle	ℓ^3	Eq. (10.23)	MAP force, particle	ℓ^3	Eq. (11.13)
DEP force, system	ℓ	Eq. (10.31)	MAP force, system	ℓ	Eq. (11.13)
Thermal diffusion time	ℓ^2	Eq. (12.16a)	Acoustic impedance	ℓ^0	Eq. (15.61)
Thermal resistance	ℓ^{-1}	Eq. (12.59)	Acoustic radiation force	ℓ^3	Eq. (15.81)
Thermal capacitance	ℓ^3	Eq. (12.61)	Optical absorbance	ℓ^1	Eq. (16.25b)
Thermal RC-time	ℓ^2	Eq. (12.63)	Optical damping coeff.	ℓ^0	Eq. (16.14)

Ref: Hendrik Bruus "Theoretical Microfluidics"

Example of flow



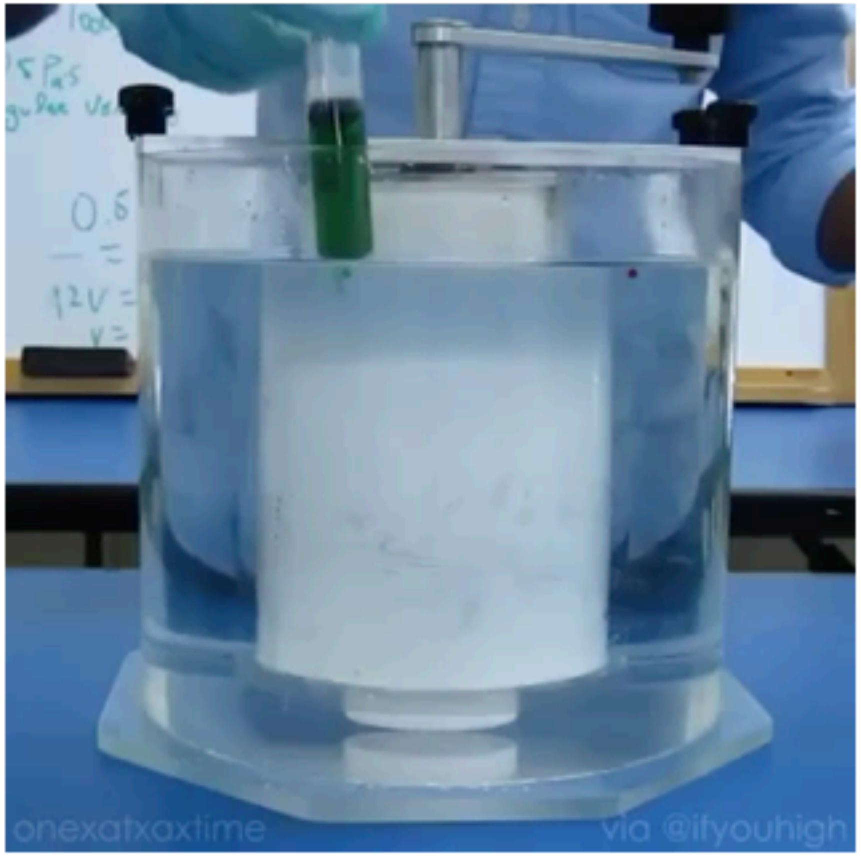
Turbulent flow



Laminar flow

Images: Wikipedia; Whitesides group

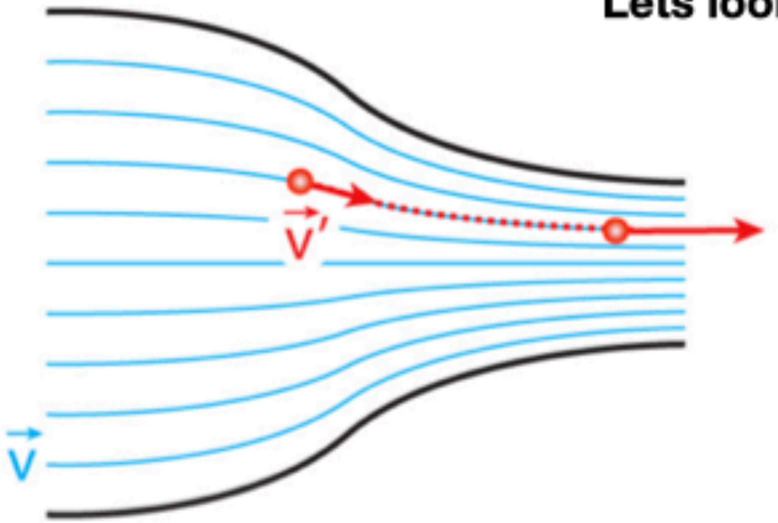
Example of flow



Video: https://www.youtube.com/watch?v=Yy0-1nWVgls

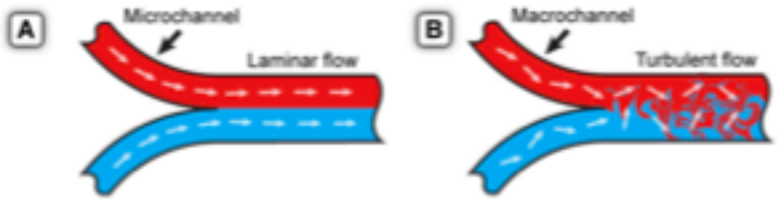
How does that come?

Lets look some fluid physics



$$\vec{r} = L_0\tilde{\vec{r}} \ ,\nabla = \frac{1}{L_0}\tilde{\nabla}, \vec{v} = v_0\tilde{\vec{v}} \ , t = \frac{L_0}{v_0}\tilde{t} \ and \ P = \frac{\eta v_0}{L_0}\tilde{P}$$

$$\underbrace{\frac{\rho v_0 L_0}{\eta}}_{Re}\left(\frac{\partial\tilde{\vec{v}}}{\partial\tilde{t}} + \tilde{\vec{v}}\cdot\tilde{\nabla}\tilde{\vec{v}}\right) = -\nabla\tilde{P} + \tilde{\nabla}^2\tilde{\vec{v}}$$



Navier-Stokes equation

$$\rho\left(\frac{\partial\vec{v}}{\partial t} + \vec{v}\cdot\nabla\vec{v}\right) = -\nabla P + \eta\nabla^2\vec{v}$$

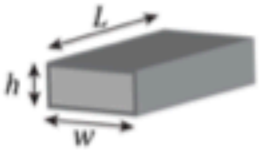
Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

Navier-Stokes for microfluidics

Low Re
Laminar flow

$$\nabla P = \eta \nabla^2 \vec{v}$$

This is simple :)



$$v_x(y,z) = \frac{4h^2 \Delta p}{\pi^3 \eta L} \sum_{n,odd}^{\infty} \frac{1}{n^3} \left[1 - \frac{\cosh\left(n\pi \frac{y}{h}\right)}{\cosh\left(n\pi \frac{w}{2h}\right)} \right] \sin\left(n\pi \frac{z}{h}\right)$$

$$v_y = v_z = 0$$

We can solve it for example for the boundary conditions of rectangular channel -> **velocity profile**

$$Q = \int_0^h \int_{-w/2}^{w/2} v_x(y,z)\,dy\,dz$$

$$= \frac{h^3 w \Delta p}{12 \eta L} \left[1 - \sum_{n,odd}^{\infty} \frac{1}{n^5} \frac{192}{\pi^5} \frac{h}{w} \tanh\left(n\pi \frac{w}{2h}\right) \right]$$

If we integrate it over the channel cross-section, we get **flow rate**

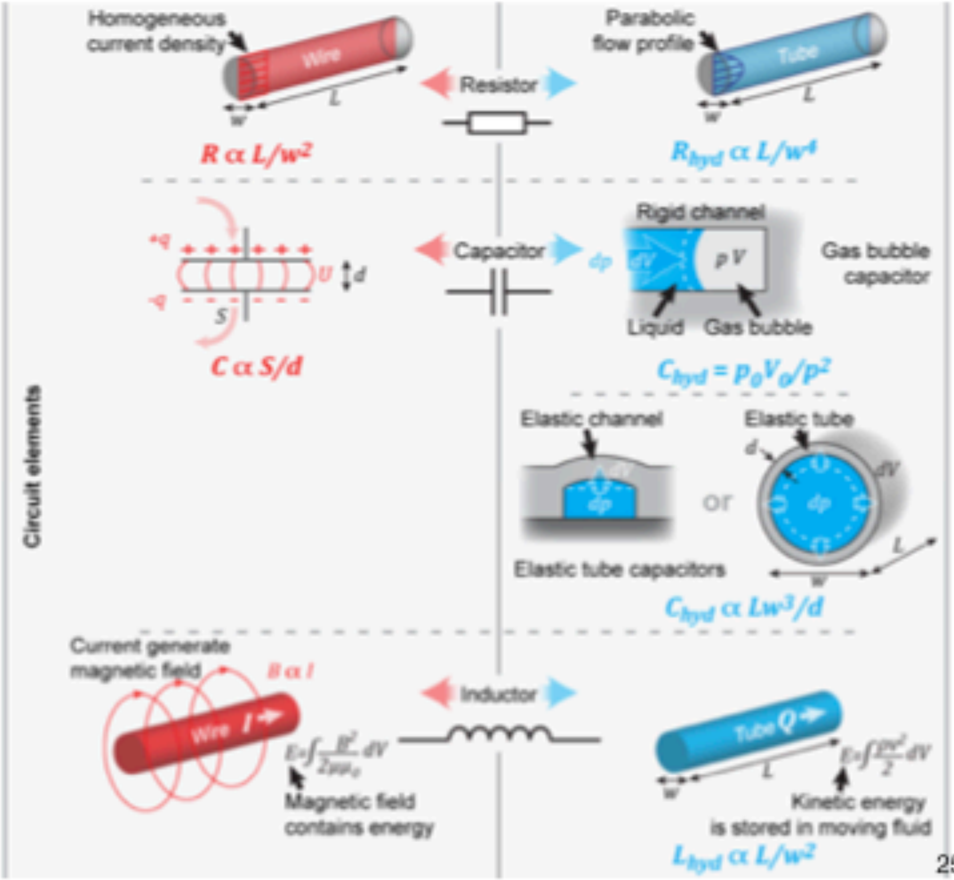
$$Q = \Delta p \frac{h^3 w}{12 \eta L} \left[1 - 0.630 \frac{h}{w} \right]$$

This is approximate formula!
Maximum error 13%
What we see here?

Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

Fluidics *vs* electronics - analogies

	Electronics		Fluidics
Flow Properties	Voltage / U (Unit: V)	Driving potential	Pressure / p (Unit: Pa)
	Charge / q (Unit: C)	Transported substance	Fluid volume / V (Unit: m³)
	Current / I (Unit: A)	Transport rate	Flow rate / Q (Unit: m³/s)
	Current density / σ (Unit: A/m²)	Transport intensity	Flow velocity / v (Unit: m/s)
Circuit properties	Resistance / R (Unit: Ω) Ohm's law $I=\frac{U}{R}$	Resistance	Hydraulic Resistance / R_{hyd} (Unit: Pa s/m³) Hagen-Poiseuille law $Q=\frac{p}{R_{hyd}}$
	Capacitance / C (Unit: F) $dq=CdU$	Capacitance	Compliance / C_{hyd} (Unit: m³/Pa) $dV=C_{hyd}\,dp$
	Inductance / L (Unit: H) $dU=-LdI/dt$	Inertia	Inertia / L_{hyd} (Unit: Pa s²/m³) $dp=-L_{hyd}\,dQ/dt$



Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

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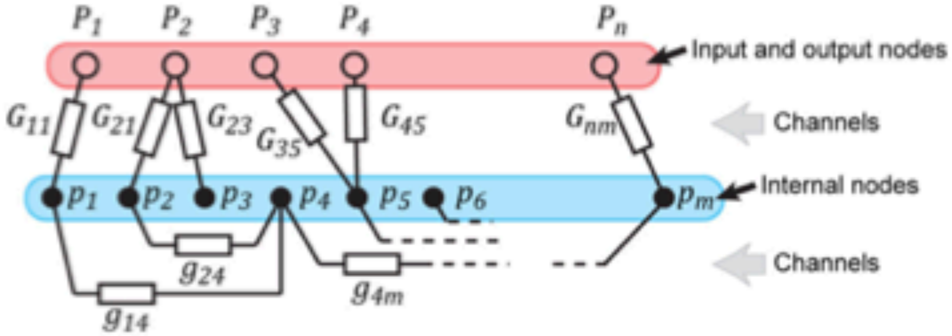
Practical flow calculations

If you have a network of channels?
What to do?
Use Kirchoff's rules and solve a linear equation system, it's like resistor network

$$\underbrace{-\left(\sum_{j=1}^{m} g_{ji} + \sum_{j=1}^{n} G_{j1}\right)p_i}_{\text{outflow from the node } i} + \underbrace{\sum_{j=1}^{m} g_{ji}p_j}_{\substack{\text{inflow from}\\ \text{other internal}\\ \text{nodes to node } i}} + \underbrace{\sum_{j=1}^{n} G_{ji}P_j}_{\substack{\text{inflow from}\\ \text{input nodes to}\\ \text{node } i}} = 0$$

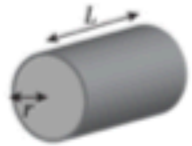
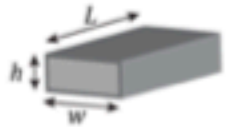
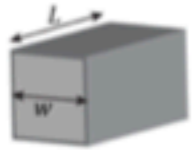


$$\underbrace{\begin{pmatrix} -\sum_{i=1}^{m} g_{i1} - \sum_{j=1}^{n} G_{j1} & \cdots & g_{1m} \\ \vdots & \ddots & \vdots \\ g_{m1} & \cdots & -\sum_{i=1}^{m} g_{im} - \sum_{j=1}^{n} G_{jm} \end{pmatrix}}_{g} \underbrace{\begin{pmatrix} p_1 \\ \vdots \\ p_m \end{pmatrix}}_{\vec{p}} = -\underbrace{\begin{pmatrix} G_{11} & \cdots & G_{n1} \\ \vdots & \ddots & \vdots \\ G_{1m} & \cdots & G_{nm} \end{pmatrix}}_{G} \underbrace{\begin{pmatrix} P_1 \\ \vdots \\ P_n \end{pmatrix}}_{\vec{P}}$$

$$\vec{p} = -g^{-1}G\vec{P}$$

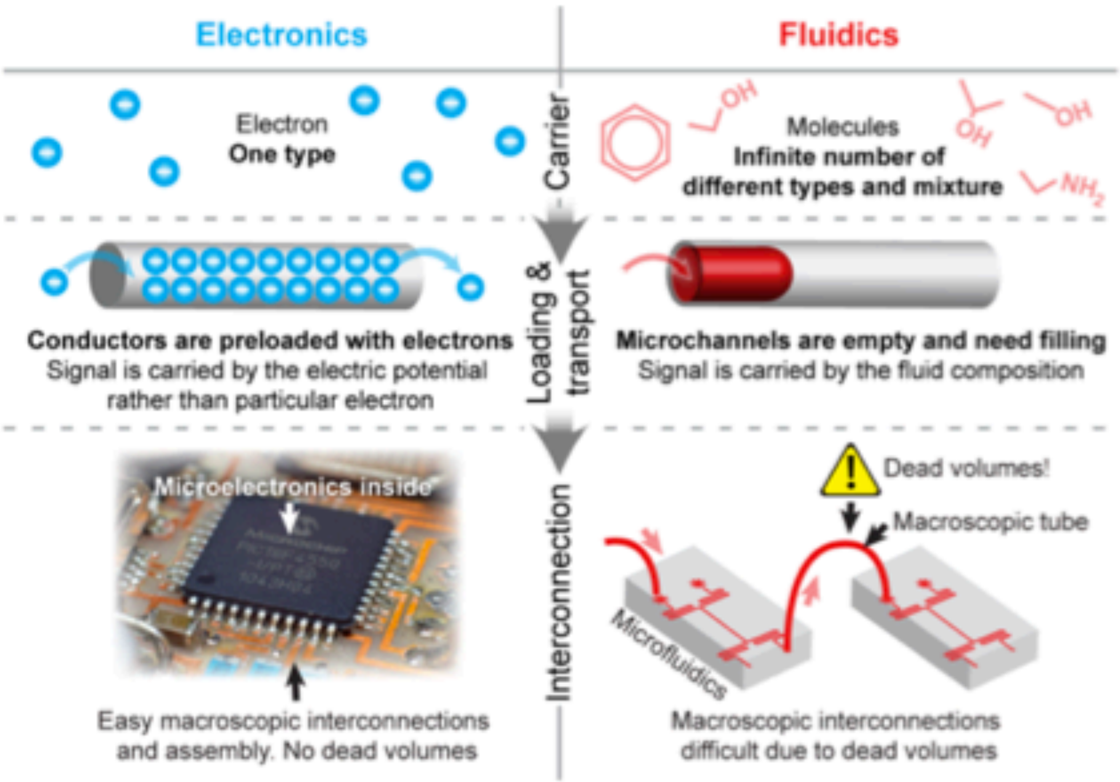
Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

Practical flow calculations

Geometry	Channel resistance	Figure
Circular	$R = \frac{8}{\pi}\eta L \frac{1}{r^4}$	
Rectangular	$R \approx \frac{12}{1-0.63(h/w)}\eta L \frac{1}{h^3 w}$	
Square	$R = 28.4\eta L \frac{1}{w^4}$	
Parabolic	$R = \frac{105}{4}\eta L \frac{1}{h^3 w}$	

Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

Fluidics *vs* electronics - differences



Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

This leads us to other transport phenomena: Diffusion

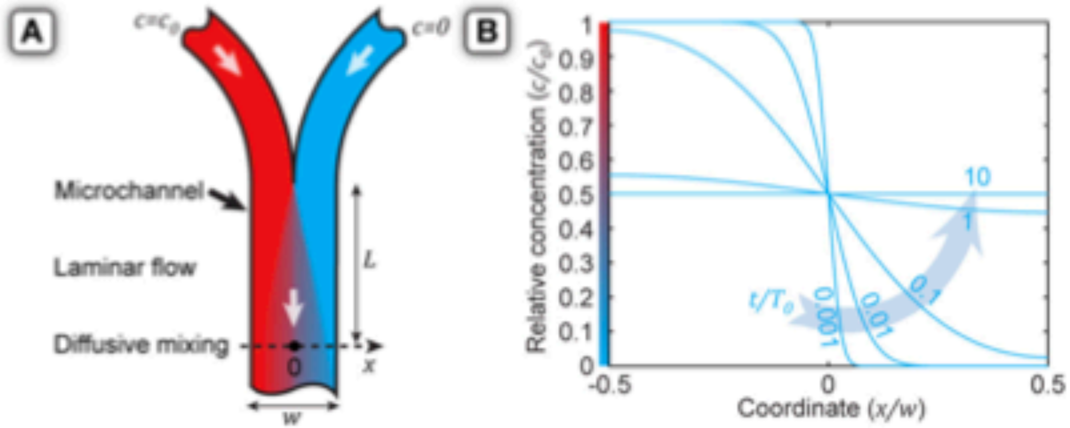
$$\frac{\partial c}{\partial t} = -\vec{v} \cdot \nabla c + D\nabla^2 c$$

Convection-diffusion equation

$$St\frac{\partial \tilde{c}}{\partial \tilde{t}} = -\vec{\tilde{v}} \cdot \tilde{\nabla}\tilde{c} + \frac{1}{Pe}\tilde{\nabla}^2\tilde{c} \qquad St = \frac{L_0}{\tau v_0} \quad and \quad Pe = \frac{v_0 L_0}{D}$$

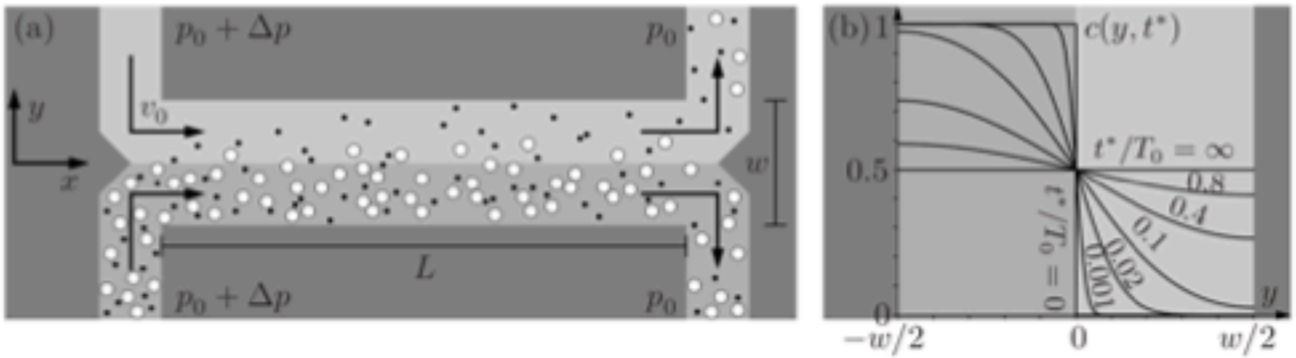
Dimensionless convection-diffusion equation and Peclet number

Steady flow



Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

Application example: H-filter



$$T_0 = \frac{L_0^2}{D},$$

Stokes-Einstein equation

$$D = \frac{k_\mathrm{B} T}{6\pi \eta r}$$

Example values

$D \approx 2 \times 10^{-9}$ m^2/s, small ions in water,

$D \approx 5 \times 10^{-10}$ m^2/s, sugar molecules in water,

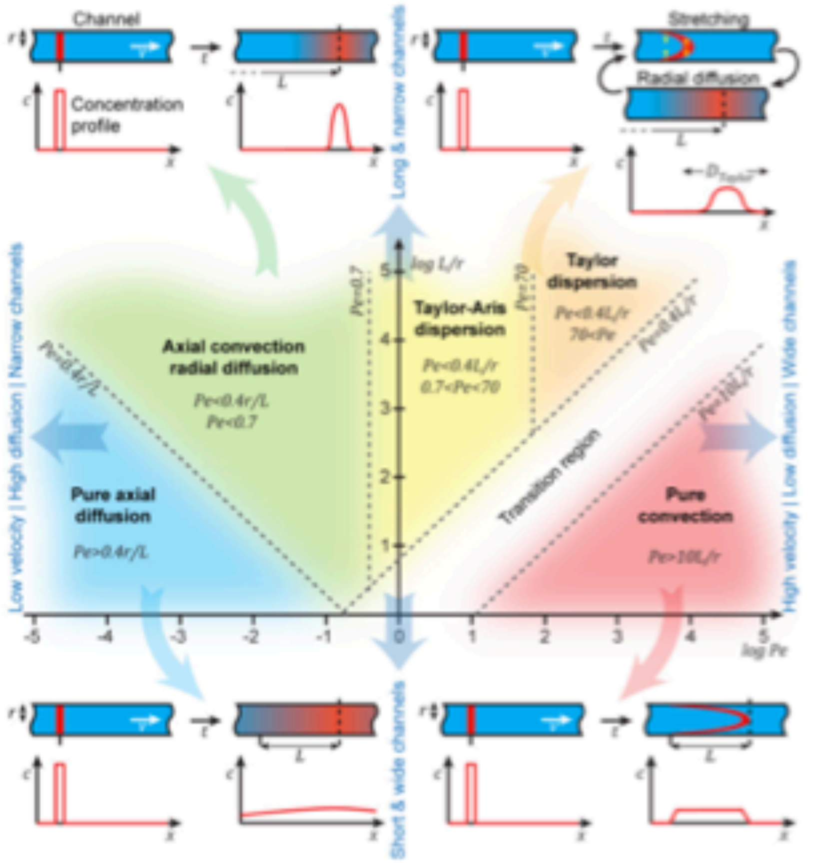
$D \approx 4 \times 10^{-11}$ m^2/s, 30-base-pair DNA molecules in water,

$D \approx 1 \times 10^{-12}$ m^2/s, 5000-base-pair DNA molecules in water,

What for ions takes to diffuse for seconds for DNA is hours!

Ref: Hendrik Bruus "Theoretical Microfluidics"

Time-dependent convection diffusion



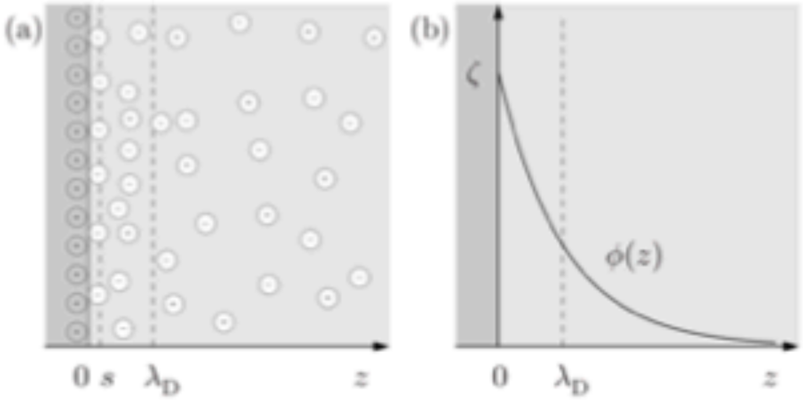
Transient flow is more complicated and there are different regimes

Ref: Alar Ainla "The Multifunctional Pipette" (PhD thesis)

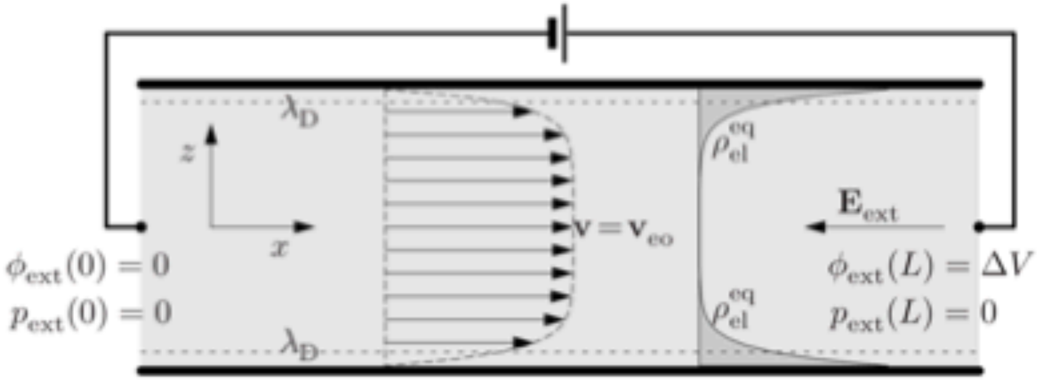
But there are even more different transport modes

By electricity: electroosmosis

On the surface of channel - there is a charge



Debye layer and
Zeta-potential



If you apply voltage
liquid will move - but bit different
from pressure driven flow!

$$v_{eo} \equiv \frac{\epsilon\zeta}{\eta} E.$$

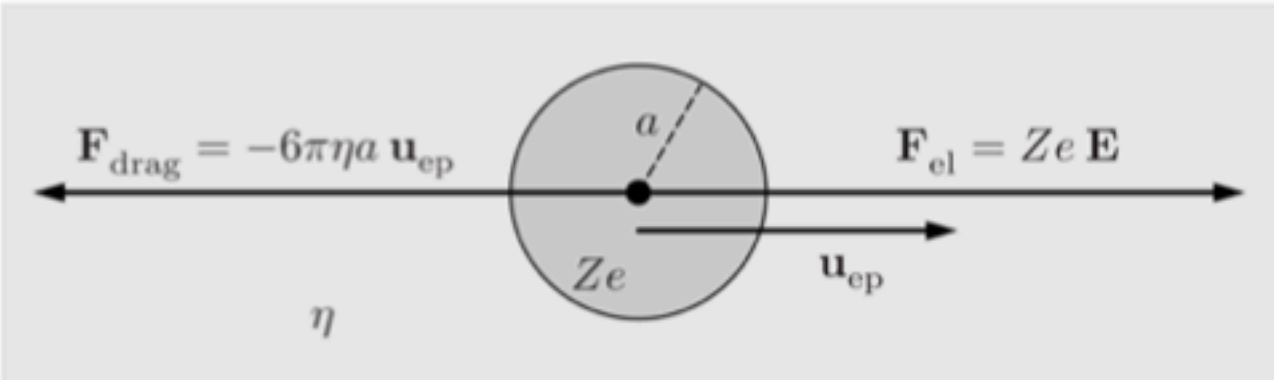
Example values

$\zeta \approx 100\ \mathrm{mV}, \quad \mu_{eo} \approx 7 \times 10^{-8}\ \mathrm{m^2\,(V\,s)^{-1}}, \quad v_{eo} \approx 1\ \mathrm{mm\,s^{-1}}.$

Ref: Hendrik Bruus "Theoretical Microfluidics"

By electricity: electrophoresis

When we have charged particles in the solution - e.g. molecules



Electric field pulls

$$\mathbf{u}_{\text{ep}} = \frac{Ze}{6\pi\eta a}\,\mathbf{E} \equiv \mu_{\text{ion}}\,\mathbf{E}$$

Viscosity holds back

Stokes-Einstein equation

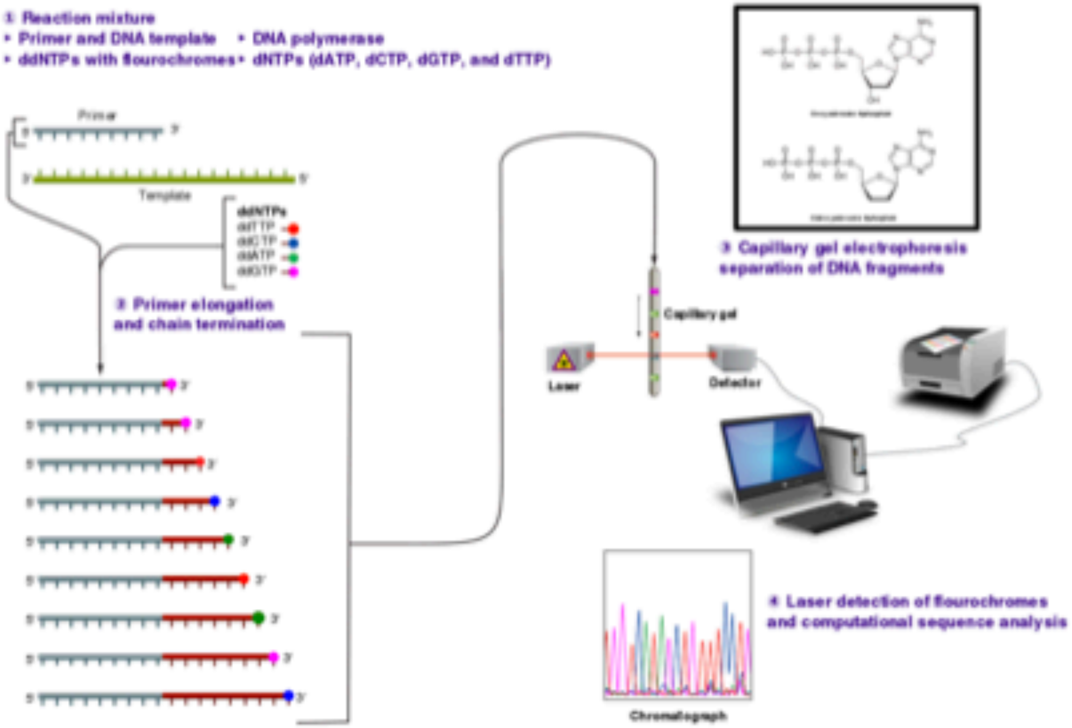
Similar

$$D = \frac{k_{\text{B}}T}{6\pi\,\eta\,r}$$

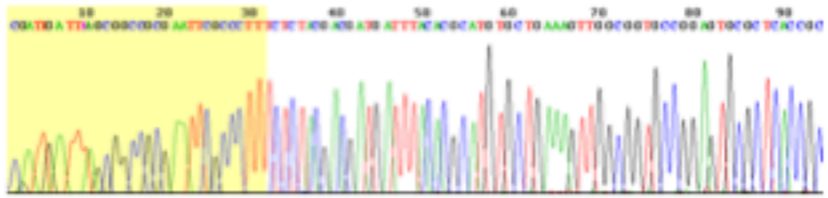
Ref: Hendrik Bruus "Theoretical Microfluidics"

Application example: Capillary electrophoresis (CE)

Sanger sequencing - separation of biomolecules



Commercial CE instrument

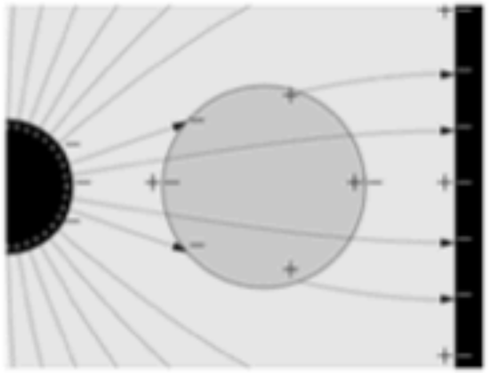


Electropherogram of DNA
shorter DNA fragments are faster

Image: CSBI@MIT and Wikipedia (Sanger Sequencing)

Electricity: dielectrophoresis

Here particles are **charge neutral** but they are moved by **inhomogeneous electric field**, because they have different polarizability (dielectric permeability) than liquid



$$U_{ES} = \frac{1}{2}\varepsilon \int_V |\mathbf{E}|^2 \,\mathrm{d}V\,,$$ Energy of electric field

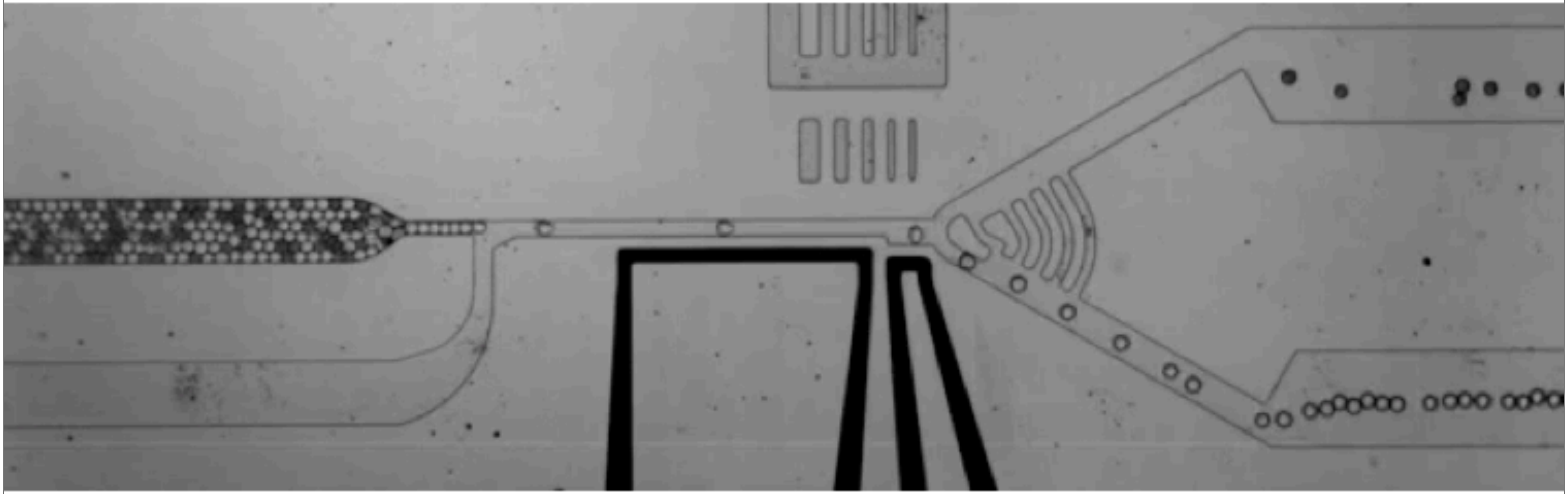
If dielectric permeability of particle is larger than the environment it is pulled toward higher field strength, otherwise it is pushed away

Field can be DC or AC
AC field can change the sign depending on the frequency!

Ref: Hendrik Bruus "Theoretical Microfluidics" & Wikipedia (Electric field)

Application example

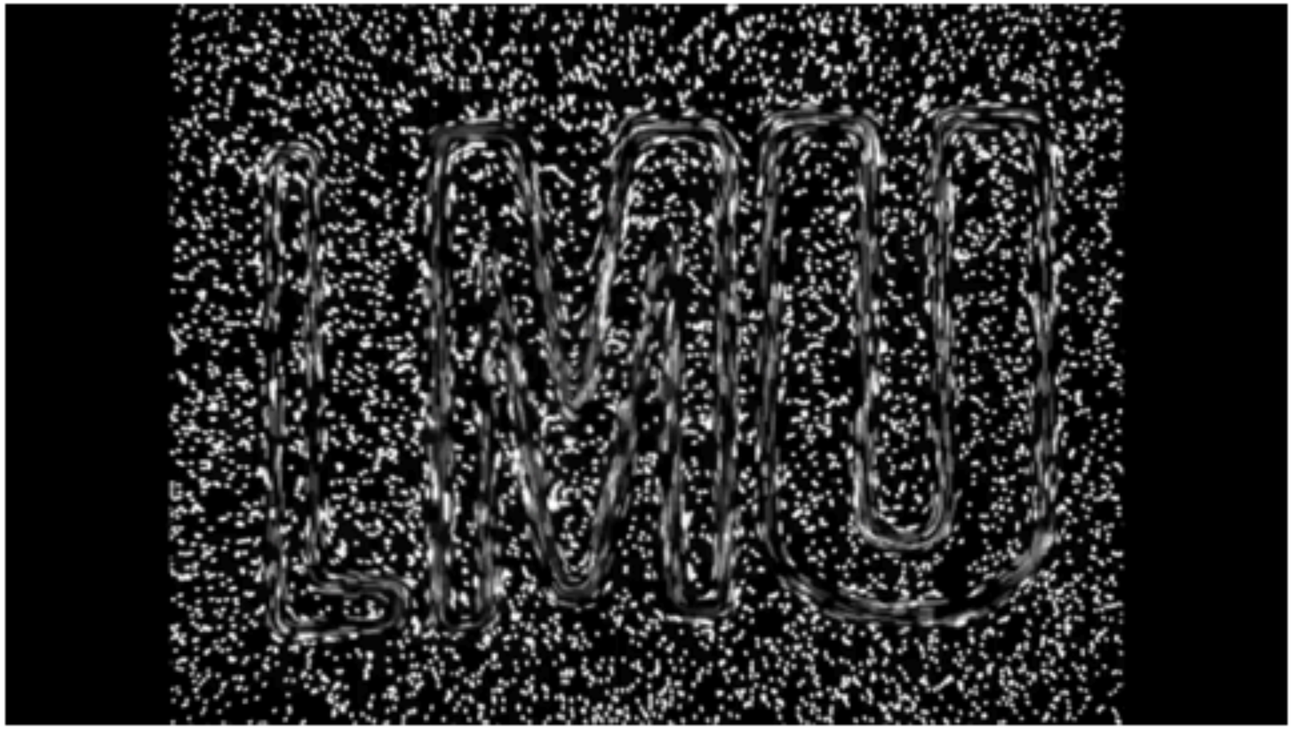
Fast sorting



Video: https://www.youtube.com/watch?v=4zsSET80jpQ & Article: Agresti et. al PNAS 2010

Thermal transport

Quiz: What is happening here?

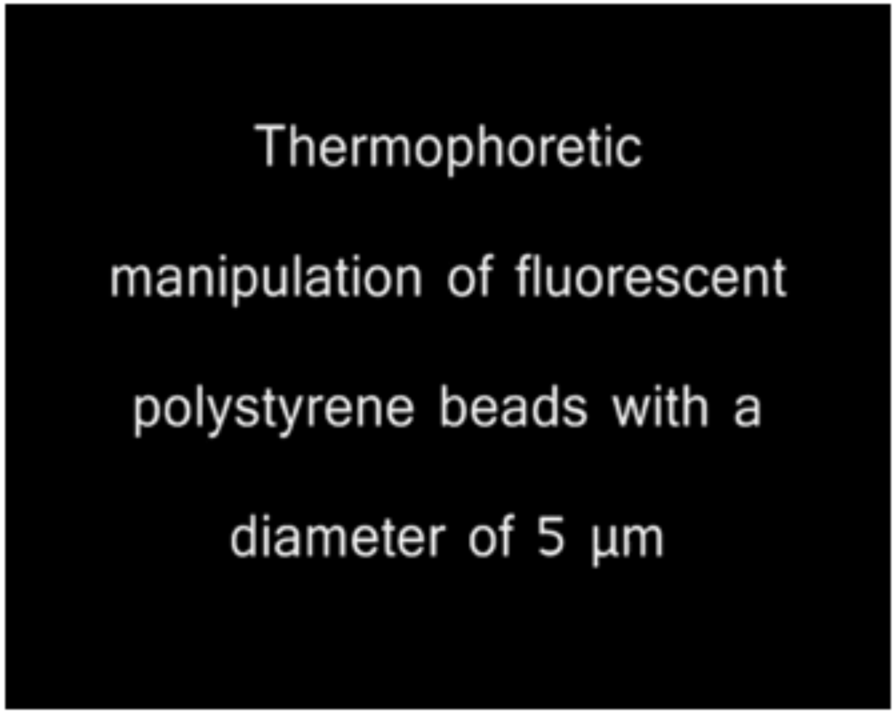


Video: https://www.youtube.com/watch?v=EGzu7pivi9w & Article: Weinert et al Phys. Rev. Lett. 2008

Thermal transport

- Temperature has number of effects:
  - It causes thermal expansion
  - It reduces viscosity of the fluids
  - It increases diffusion
  - Thermophoresis
  - In cases of chemical reactions, it accelerates reactions
  - Temperature itself propagates similarly to diffusion, only much faster and through different materials

Thermophoresis



Video: https://www.youtube.com/watch?v=HSDAbjzyAnw

Surfaces

Scaling -
going smaller makes surface effects more dominant over volume effects

Surfaces

We already saw a little bit about surface charge —> EOF

Surfaces have many properties, most important are:
* chemistry - chemical groups which can react and interact with the liquid
* Surface micro- and nano-structuring - smooth, porous, fibrous, rough
* These can lead to some macroscopic effects, like contact angle

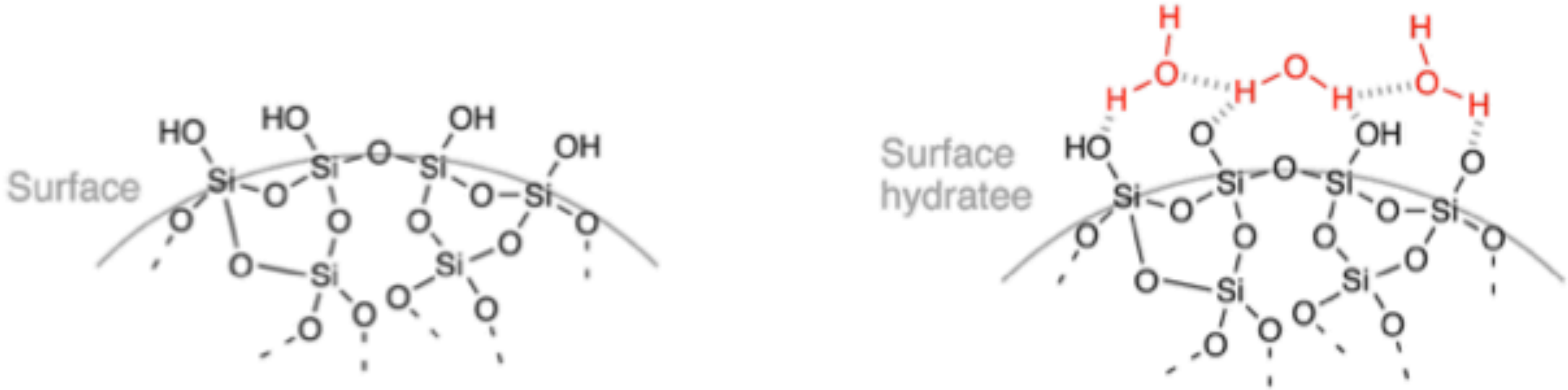
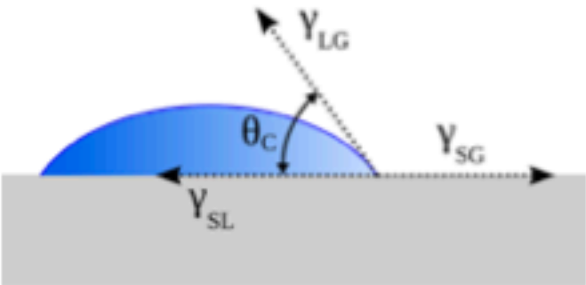


Image: Wikipedia

Contact angle

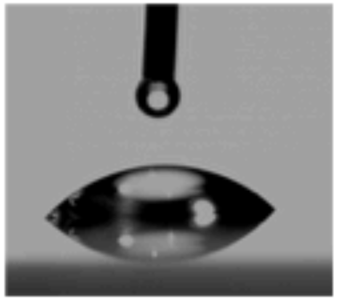
Contact angle describes the surface wettability

* Contact angle depends on the interfacial energies of three phases
* For each material combination there is unique contact angle
* Practical contact angle can be more complicated and depends on also micro-structuring (leading to contact angle hysterisis, super hydrophobicity and hydrophilicity etc)

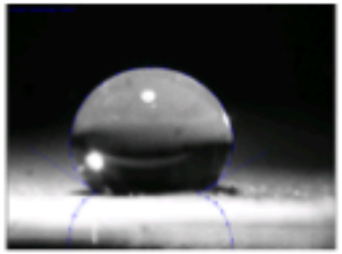


Young equation

$$\cos\theta = \frac{\gamma_{\mathrm{sg}} - \gamma_{\mathrm{sl}}}{\gamma_{\mathrm{lg}}}$$



Hydrophilic (water-glass)



Hydrophobic (water-lotus leaf)

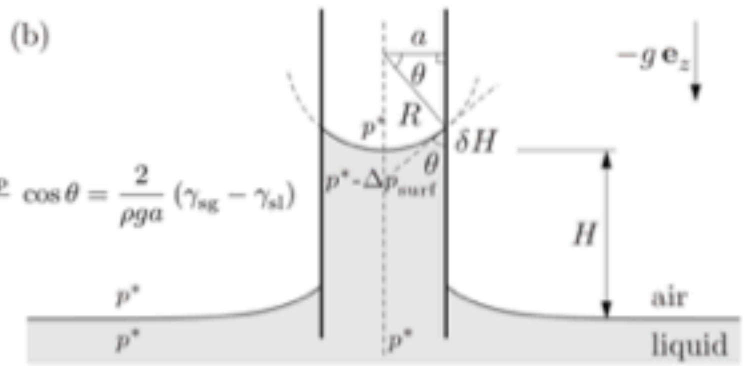
Images: Wikipedia (Contact angle)

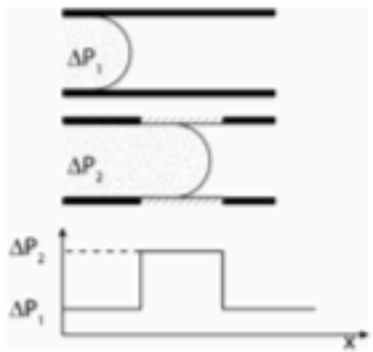
Capillary flow

Capillary effects are important in microfluidics systems

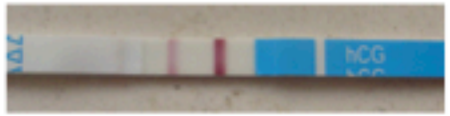
* Filling channels
  * Capillary pump (sucks liquid)
  * Capillary valves
* Porous materials
  * Wicking liquids

$$H = \frac{2\gamma}{\rho g a}\cos\theta = 2\,\frac{\ell_{cap}^2}{a}\cos\theta = \frac{2}{\rho g a}\left(\gamma_{sg} - \gamma_{sl}\right)$$





Capillary valve



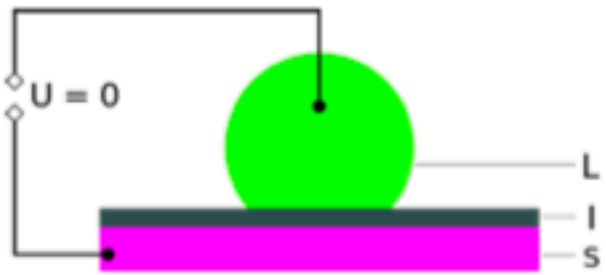
Pregnancy test

Water strider

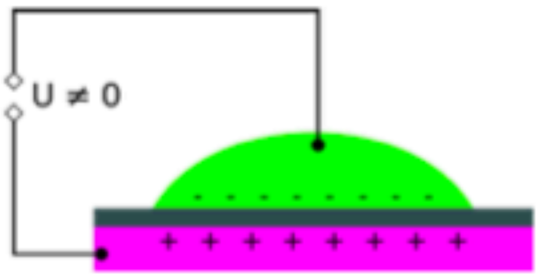
Ref: Hendrik Bruus "Theoretical Microfluidics" & Wikipedia (Gerridae, hGC pregnancy test strip) & Encycloperia of Microfluidics and Nanofluidics

Electrowetting

Contact angle can be controlled by electrical potential



Hydrophobic



Hydrophilic

$$\cos\theta = \left(\frac{\gamma_s - \gamma_{ws}^0 + \frac{CV^2}{2}}{\gamma_w} \right)$$

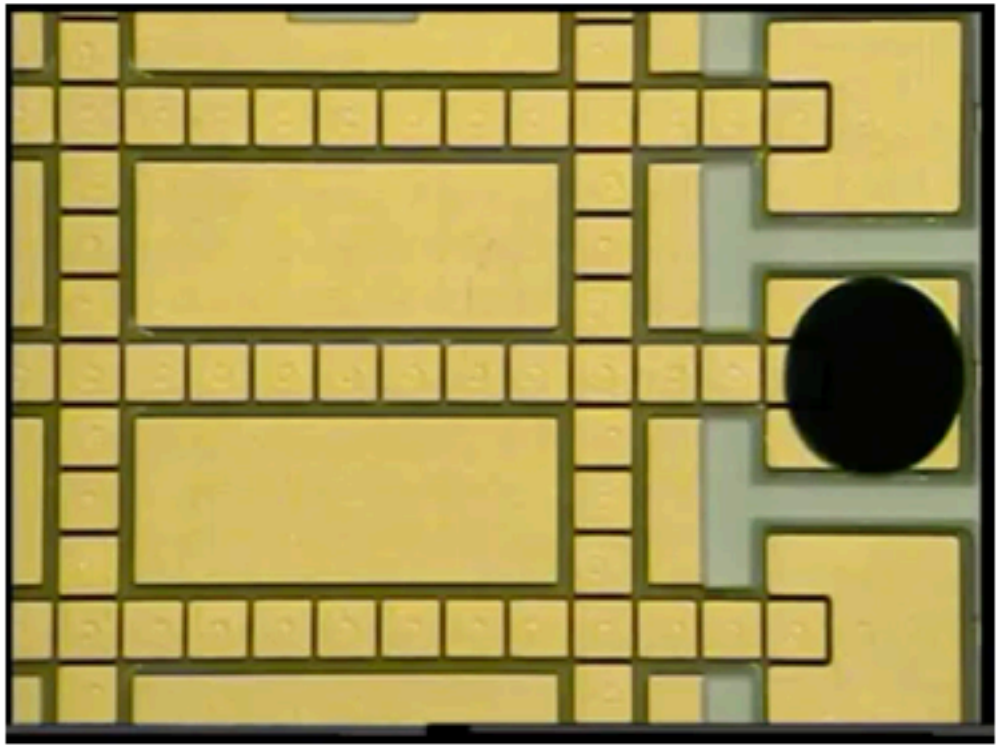
Young-Lippmann equation

EWOD - Electrowetting on dielectric

Images: Wikipedia (Electrowetting)

Applications: digital microfluidics

Manipulation on sample drops



Video: Baebies Inc, https://www.youtube.com/watch?v=fqS-TyuplEk

Applications: Electrowetting in textile microfluidics

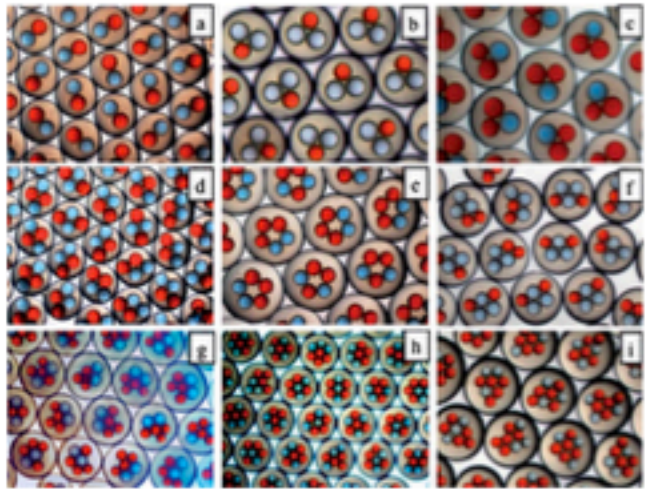
Video: Ainla et al, Advanced Materials, 2017

Interface between immiscible liquids - droplet microfluidics

Liquid-liquid interface - droplets & plugs

It's bit like a dripping water tap! But at microscale!

Due to instability interface between immiscible liquids (e.g. water in oil), they try to minimize surface energy and break into droplets and plugs, due to laminar flow regime process is extremely repeatable



Large variety of technologies to make, manipulate and handle advanced droplets exists

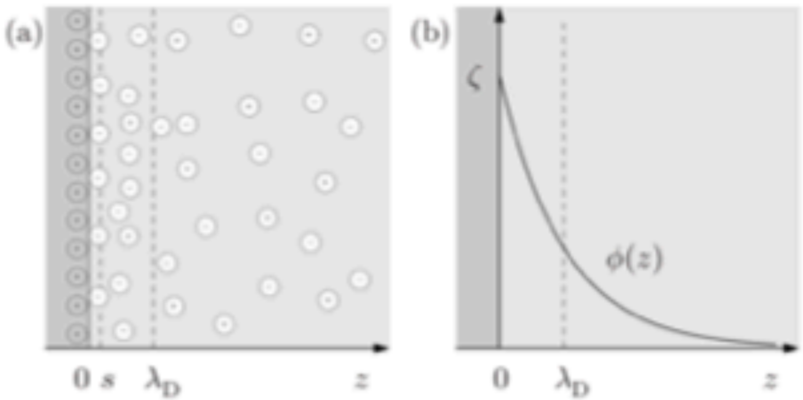
Images: Elveflow, frankelab.de

Examples

Video: Cidar Lab, https://www.youtube.com/watch?v=V_Vs81pLYoQ

Even smaller - nanofluidics

Liquid solid surfaces are not really discrete - they are bit fuzzy, what happens if channels are so small?



When does the channel becomes nano?
Depends on solution

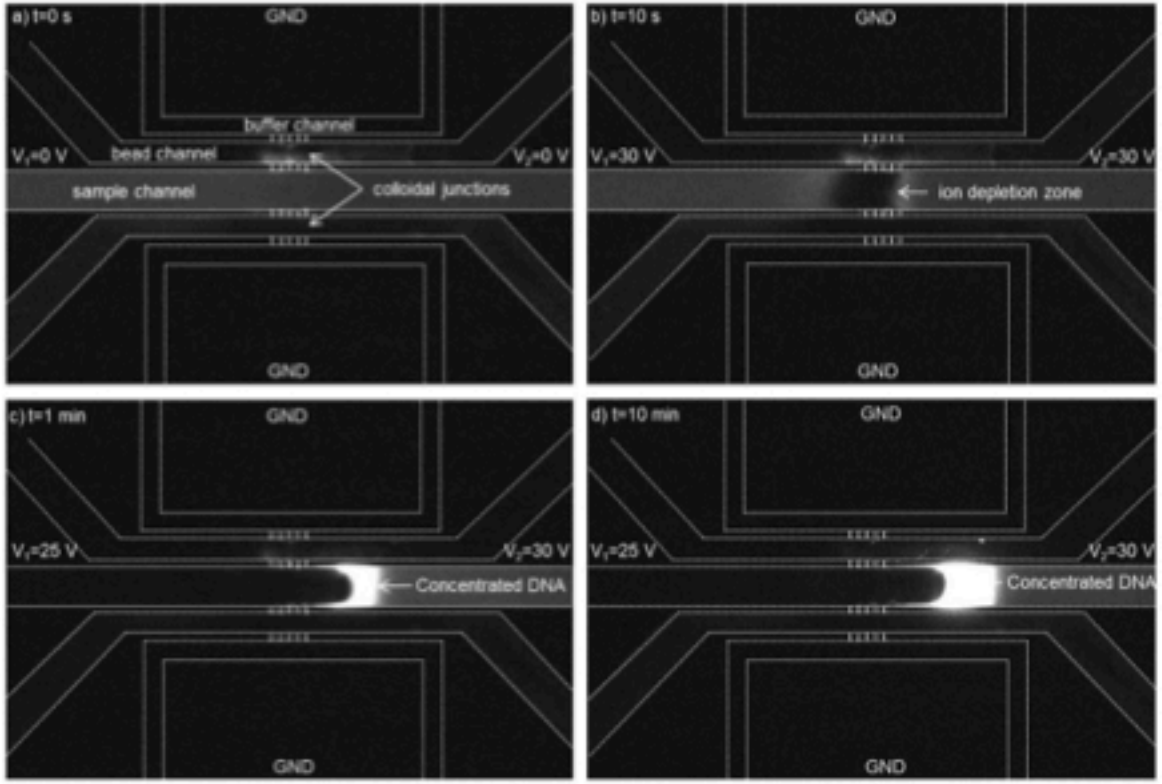
$$\lambda_D \approx \sqrt{\frac{1\ \mathrm{mM}}{Z^2\, c_0}} \times 9.6\ \mathrm{nm}$$

Human serum 150mM —> 0.8nm (very small)
1mM solution —> 9.6nm
10µM solution —> 96nm
deionized water —> can be up to ~1µm

Ref: Hendrik Bruus "Theoretical Microfluidics"

Example

Such channels can filter molecules and ions by charge



Ref: https://www.jove.com/video/54145/creating-sub-50-nm-nanofluidic-junctions-pdms-microfluidic-chip-via

Summary … what did we talk today

- What is Microfluidics?
- Brief history of Microfluidics?
- Why Microfluidics? Why should we care about it?
- Scaling laws - why “micro” is different?
- Different transport phenomena
  - Flow (Flows on microscale & some practical calculations)
  - Diffusion
  - Flows & transport driven by electricity
    - Electroosmosis
    - Electrophoresis
    - Dielectrophoresis
  - Thermally driven flow (thermophoresis)
- Surfaces
  - Capillary effect
  - Electrowetting
  - Liquid-liquid interfaces and droplets
- Going even smaller - nanofluidics

Don’t forget homework!

Hope you learned something new about microfluidics :)
Thank you!